

Senior Software Engineer Case Study

Project & Resource Management System

Objective

Design and develop a web-based Project & Resource Management System for a consulting firm that helps manage projects, consultants, clients, and project tasks.

The solution should demonstrate not only your ability to develop a working application, but also your software engineering practices, technical decision making, scalability considerations, and code quality.

Scenario

Build a system to manage clients, projects, consultants, project tasks and timesheets.

Functional Requirements

- Authentication & Role-Based Access (Administrator, Project Manager, Consultant)
- CRUD for Clients, Projects, Consultants, Tasks and Timesheets
- Dashboard with project and utilization metrics
- Search, filtering, sorting and pagination
- REST APIs using Django REST Framework
- Validation, logging and error handling
- Responsive Bootstrap interface

Technical Requirements

- Django 5.x
- PostgreSQL (SQLite acceptable for demo)
- Git source control
- Clean architecture and coding standards
- README with setup instructions

Bonus Features

Frontend

- React.js or Vue.js frontend consuming Django REST APIs.
- Responsive and polished UI/UX.
- Dark Mode

Backend

- JWT Authentication.
- Role-based permissions.
- Audit Trail / Activity Logging.
- File Uploads.
- Email Notifications.
- Background Jobs (Celery).

Engineering Practices

- Docker.
- CI/CD pipeline.
- Unit or Integration Tests.
- Swagger / OpenAPI Documentation.
- Redis Caching.
- Azure Deployment.

Deliverables

1. GitHub repository
2. Sample data
3. Architecture Diagram and ERD (max 5 pages)
4. 20–30 minute presentation covering architecture, code walkthrough, design decisions and future improvements

Interview Exercise

Candidates will be given a new business requirement during the interview and must explain how they would evolve the solution without coding.